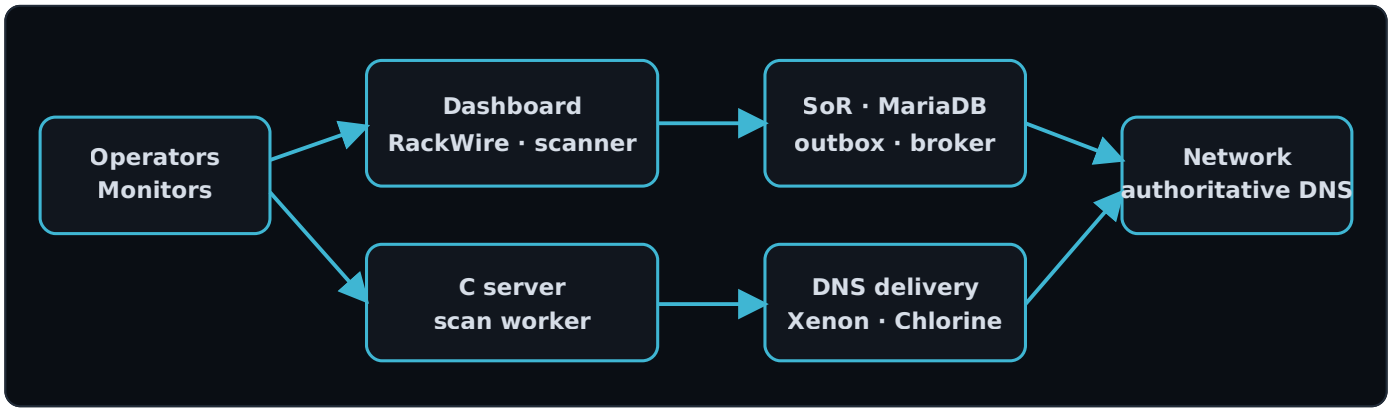
Status & Operations Guide

SolariNet platform · operator / developer handbook · current status 22 October 2026

This guide summarizes system operation and retains the Phase 7 implementation notes below. It is self-contained with inline CSS, JavaScript, and SVG. The dated status card is authoritative for the current integration candidate; Phase 7 counts and claims below are historical snapshots. Follow the linked stabilization workflow and baseline for current release gates.

Current platform state · 22 October 2026

The integration candidate is not released: as of this snapshot, the local branch is 31 commits ahead of Forgejo and GitHub `main`, whose refs match at `5e8d688`. The worktree is clean. The public review branch is open as GitHub draft PR `#37`; no main branch merge or release tag exists. The Xenon backup utility is deployed as a local-only recovery path.



- **RabbitMQ:** Benzene runs the broker in Podman on RabbitMQ `3.13.7`. It reports fully booted with no local alarms. The SoR apply queues for AD, Pi-hole, and DNS each had one running consumer and zero ready or unacknowledged messages; notify dispatch also had one consumer and no backlog. Xenon's emitter, apply workers, reconciliation, and notification services were active during the 22 October check. RabbitMQ was fully booted with no local alarms; its `solari` vhost had zero ready/unacknowledged messages and one consumer on `sor.apply.ad`, `sor.apply.pihole`, `sor.apply.dns`, and `notify.dispatch`. The broker is operational, but its installed release is old and upgrade/recovery remain release gates.
- **Backups:** Cesium's 21 October SoR dump restored successfully into a disposable MariaDB instance and all 41 tables passed checks. On 22 October, the Xenon runner wrote a 56.4 MB root-only archive and the exact artifact restored into scratch MariaDB; all 29 base tables passed checks. Xenon's nightly timer is enabled for a local root-NVMe target. This shares

Xenon's host failure domain. NAS-X provisioning, permissions, and an off-host restore remain unverified.

- **DNS:** Khadas VIM4 chlorine is active as an authoritative-only secondary for 109 zones. Steel at `10.0.0.11` is unreachable; its ZimaBlade tertiary DNS role awaits console redeployment.
- **Network discovery:** operators can start a bounded asynchronous scan; the native worker records live results as reviewable Discovery candidates, which remain separate from physical inventory until explicitly handled. The dashboard polls worker progress every second and refreshes candidate rows every 2.5 seconds while a scan runs, so new hosts appear before completion. RackWire launches the same live Discovery workflow and receives candidate handoff. The native-worker browser acceptance scanned the directly connected `10.0.0.0/24`, processed all 254 addresses, and recorded 14 candidates in a disposable database; the local listener appeared in Discovery before scan completion. The accepted path polls live worker status, refreshes candidates during execution, and supports reviewing and ingesting discovered candidates into inventory. No production database was changed. A separate UI acceptance verified candidate refresh before its controlled scan fixture reaches a terminal state. Larger-range throughput and deployed worker behavior remain unverified. Adoption mirrors a host and address into monitoring and SoR; it does not create a physical asset.
- **Inventory:** scanner export, dashboard import, and transactional SoR paths are implemented and pass scratch-database tests. No scanner rows have been imported into the live SoR. The launched scanner app's actual two-item/two-asset export was previewed, applied, and replayed in a disposable SoR; counts and location readback passed. Source locations "Bin #5" and "Nook" do not identify which of the two duplicate root "Painted Fern" sites (IDs `9103`, `9104`) is correct. The standalone scanner-ingest app has a local Git repository but no configured release remotes; GitHub and Forgejo lookups found no published `scanner-ingest` repository, so its release ownership/path is unresolved.
- **UniFi topology:** a new GET-only discovery command now reads all network pages and details, validates gateway/subnet/DHCP relationships, and emits a canonical snapshot without writing to UniFi or the SoR. The live Default site currently reports only VLAN 1, `10.0.0.0/8`, gateway `10.0.0.1`, and DHCP `10.1.0.26–10.7.161.117`. This conflicts with the project's segmented network model. The command requires HTTPS and either a trusted controller CA or an explicit insecure-TLS opt-in. No topology import or SoR write is enabled; reconcile this source state before designing the transactional importer. A read-only SoR query returned 16 active /24s across eight named segments; all have VLAN and gateway unset. The UniFi /8 overlaps all 16 and matches none exactly. The cause and source authority are unresolved.

- **Release gates:** authenticated browser acceptance now passes against disposable SoR and broker fixtures, including scanner import and RackWire iframe save. Staged production browser verification, cross-lab review, production migration/rollback, and application deployment remain open. The root review branch has a GitHub draft PR `#37`; RackWire's public GitHub mirror now has main and integration branches plus draft PR `#1`. Its GitHub mirror was created because root CI recursively checks out the RackWire gitlink. Scanner-ingest at `~/Code/scanner-ingest` remains a local repository without remotes or matching GitHub/Forgejo repositories. Claude CLI reports no authenticated session, so cross-lab review is unavailable. Root PR CI is being used to expose further issues; run `39701235974` at `686d802` passed 14 of 15 jobs, including realtime scan and candidate-ingest browser acceptance. Its remaining clean-install schema comparison still fails in hosted CI, although an equivalent local MariaDB 11.8.9 export matches the snapshot. The exporter now canonicalizes the version-specific executable-comment line and CI prints a unified diff for the next run. Earlier runs also found a date-sensitive DNS assertion and checkout/artifact wiring defects. Keep the PRs in draft until all checks and independent review pass.

Current evidence and next gates: Stabilization Baseline and Integration Workflow.

Contents · Phase 7 deliverables · Build matrix & options · Running server / client / monitor · Dashboard on-prem setup · API / control-plane discipline · The MCP · Enrollment runbook · Decommission runbook · Retention · CI · Known gaps

Phase 7 deliverables

Topology — deep path (server)

`src/server/serverTopology.c` gained a deep-discovery path that normalizes LLDP-MIB rows into the schema's gear/edge model and assembles the `/api/netgear` view — all pure and SNMP-free in the default build:

- **LLDP-MIB normalization.** One decoded `lldpRemTable` row (`topoLldpMibRecord`) becomes a `serverNetGear` + a `serverLldpEdge` (`topoNormalizeMibRecord`). Gear-kind inference (`topoInferGearKind`) treats the LLDP `SystemCapabilitiesMap` bits as authoritative (AP > router > bridge), falling back to sysName / sysDesc keyword matching when caps are absent. Gear IDs (`topoMibGearId`) are stable and kind-prefixed (`sw-/ap-/gw-/rtr-/nd-`), seeded sysName

→ mgmtIp → chassisId. A C99-safe case-insensitive substring (`topoContainsCi`) avoids GNU `strcasestr` .

- **Uplink-chain + attached-count assembly.** `topoAssembleNetGear` de-duplicates gear by gearId, counts DISTINCT attached SolariNet nodes per gear (gear-to-gear edges excluded; a node seen on multiple interfaces counts once), and computes uplink depth with a cycle guard (`TOPO_MAX_UPLINK_HOPS`). Returns `ERR_BUFFER_FULL` on an undersized output buffer.
- **Two projections.** `/api/topology` serves `view=monitoring` (from HRW + node data) and `view=network` (from `networkGear` + `lldpEdge` + the assembled uplink chain).

> **SNMP capabilities are optional.** With `SOLARI_WITH_SNMP=ON` , the LLDP topology fetch (`topoSnmpFetchLldp`) walks LLDP-MIB columns with the net-snmp CLI, then feeds the tested normalizer; the IF-MIB poller stores interface counters in `gearInterfaceCurrent` and `gearInterfaceHistory` . Both fetch paths return `ERR_PLATFORM` in the default build. The option is off by default and adds no SNMP library build dependency, but requires `snmpbulkwalk` at runtime. Neither path has been verified against a live network device in this release candidate. The IF-MIB walk parser now rejects malformed numeric OIDs, invalid or out-of-range values, and interface names that would otherwise be silently truncated; tests pass with the feature both off and on. A fake CLI integration fixture exercises LLDP column collection, normalization, and nonzero-command failure handling without contacting a device.

MCP — Phase 7 catch-up

The Python / FastMCP server (`integrations/mcp/`) grew from 14 to **26 registered tools**, covering the §5/§6 read surface and a gated, audited control surface. The SQL-query tool also gained a table allow-list. Details in the MCP section.

Operational hardening

Enrollment / decommission scripts (`deploy/enrollment/`), hardened systemd units (`deploy/systemd/`), DB maintenance procedures (`db/maintenance/`), and a GitHub Actions CI workflow (`.github/workflows/ci.yml`). All additive — no C sources or dashboard files were modified to land them.

Build matrix & options

The build is layered: a zero-dependency ISO C99 core that builds anywhere, with heavier capability behind explicit CMake options. All options default `OFF` .

OPTION	DEFAULT	PULLS IN	ENABLES
`SOLARI_BUILD_TESTS`	OFF	(vendored Unity)	the ctest suites
`SOLARI_WITH_IO`	OFF	nng + mbedTLS	the transport layer (`solariNet`)
`SOLARI_WITH_SQLITE`	OFF	SQLite	the store-and-forward spool
`SOLARI_WITH_JSON`	OFF	cJSON	the JSON config overlay
`SOLARI_BUILD_SERVER`	OFF	libmariadb (+ IO transitively)	solariServer (x86_64 / arm64 only)
`SOLARI_WITH_SNMP`	OFF	net-snmp `snmpbulkwalk` at runtime (none vendored or linked)	IF-MIB interface metrics and LLDP topology walks

Cross / platform toolchains

`cmake/toolchains/` carries `linux-x86_64`, `linux-arm64`, `linux-arm32`, `windows-x86_64`, and `macos-universal`. The server tier is built only on the 64-bit Linux targets; the edge agents build across the matrix.

```
# core only (no deps, builds anywhere)
cmake -S . -B build -DSOLARI_BUILD_TESTS=ON
cmake --build build && ctest --test-dir build --output-on-failure

# full server tier
cmake -S . -B build -DSOLARI_BUILD_TESTS=ON -DSOLARI_BUILD_SERVER=ON \
      -DSOLARI_WITH_IO=ON -DSOLARI_WITH_SQLITE=ON -DSOLARI_WITH_JSON=ON
cmake --build build && ctest --test-dir build --output-on-failure
```

> **Verified in this session.** A clean-from-scratch full-server configure + build under `-Wall -Wextra -Wpedantic` reports **0 warnings, 0 errors**; MariaDB was found via pkg-config and linked. `ctest` : **0 failed across 33 registered tests** (server suites included); the DB-live test self-skips without its scratch-DB settings. The MCP suite reports **81 passed**.

Running server / client / monitor

The supported way to run the three roles is via the hardened systemd units in `deploy/systemd/` (§15.2):

UNIT	ROLE	HARDENING NOTE
`solari-server.service`	solariServer (primary or failover — same binary, lease-decided)	`RuntimeDirectory` provides the `solariCtl` socket dir
`solari-monitor.service`	probe monitor	adds `CAP_NET_RAW` for ICMP probing
`solari-client.service`	edge agent	least-privilege sandbox
`solari-client-watchdog.service`	optional external supervisor of the client-spawned watchdog sibling	defense-in-depth for the §7.3 watchdog

```
install -m0644 deploy/systemd/*.service /etc/systemd/system/
systemctl daemon-reload
systemctl enable --now solari-server.service       # the C2 tier
systemctl enable --now solari-monitor.service      # probe monitor(s)
systemctl enable --now solari-client.service       # on each monitored host
```

The candidate contains 49 service, timer, socket, and path unit files, including 43 services. The dedicated section-placement test passes across all 43 service files. `systemd-analyze verify` reports 14 missing-executable diagnostics for deployment paths absent from this development checkout; it exits nonzero for those unresolved paths. No directive-section errors were emitted. Start-limit directives on the SoR sync and notification units are in `[Unit]`, where systemd applies them.

Dashboard on-prem setup

The dashboard (Phase 6) is fully on-prem: React / ReactDOM / Babel are vendored under `dashboard/public/vendor/`, fonts and icons are local, and the page makes no external fetches. Serve `dashboard/public/` as the static root and `dashboard/api/` (a PDO-backed PHP service) under `/api` on the same origin. The API needs a PDO DSN to the server's MariaDB (reads) and access to the `solariCtl` Unix socket (control actions). See `PHASE6_REPORT.html` for the route map.

API / control-plane discipline

There are three operator surfaces — the dashboard PHP API, the MCP, and the `solariCtl` Unix socket — and they share one discipline:

- **Reads are direct, control is routed.** Read endpoints project the database. State-changing actions never write the DB from the edge surface; they are handed to the C server (via `solariCtl`), which stays the single arbiter.
- **Destructive actions require an operator token *and* explicit confirmation.** Approve-enrollment (signs a CSR) and decommission (wipes a node) refuse before any network call unless both the operator token and an explicit `confirm=true` are present. Decommission uses a stored SHA-256 digest of a random one-time confirmation token, bound to node, wipe scope, operator, and a ten-minute expiry. The server consumes it and retires the node in one transaction; raw tokens are never written to audit details.
- **solariCtl verbs:** `PING`, `SURVEY`, `APPROVE` / `REJECT`, `PROVISION`, `ADOPT`, `IGNORE`, `CONTROL`, `DECOMMISSION` / `RETIRE`.

The MCP

`integrations/mcp/` is a Python / FastMCP server exposing SolariNet to an AI assistant. It wraps the same REST surface (envelope-unwrapping `RestClient`) and fixed read-only database tools; generic caller-supplied SQL is removed. Clients are injectable for tests. Deps live in a git-ignored `.venv`. **26 tools** are listed by the current candidate CLI.

Read tools (GET, read-only)

Among the §6 reads added this phase: `solarinet_list_discovery`, `solarinet_list_enrollments`, `solarinet_list_builds`, `solarinet_list_segments`, `solarinet_list_netgear`, `solarinet_get_topology_view` (dual-hierarchy `view=monitoring|network`), and a read-only `solarinet_get_config`.

Control tools

- **Non-destructive (POST):** `solarinet_adopt_discovery`, `solarinet_ignore_discovery`, `solarinet_reject_enrollment`, `solarinet_provision` (requires node_id or enr_id).
- **Destructive (gated):** `solarinet_approve_enrollment` (signs CSR) and `solarinet_decommission` (wipe). Both are marked `destructiveHint` and require the operator

token *and* `confirm=true`, refusing before any network call — mirroring the dashboard discipline. A `_require_operator_token()` helper enforces the token gate.

Database access boundary

MCP callers cannot provide SQL. Three fixed database tools use internal queries; the DB client enforces an explicit table allow-list (`solarinet_mcp/db.py`) alongside SELECT/WITH-only, anti-stacking, no-OUTFILE checks. `_enforce_table_allow_list()` extracts every `FROM` / `JOIN` target, exempts CTE names declared via `WITH`, and rejects anything unlisted — blocking `information_schema` / `mysql` / `performance_schema` exfiltration. The allow-list covers the §10 base tables plus the new §5 tables (`segment`, `networkGear`, `lldpEdge`, `discovered`, `enrollment`, `buildArtifact`).

```
# run the MCP test suite
cd integrations/mcp && .venv/bin/python -m pytest -q     # 81 passed
```

Enrollment runbook

`deploy/enrollment/solari-enroll.sh` enrolls a node end-to-end, generating key material *locally* and only ever sending a CSR:

1. Generates an EC P-256 key + CSR locally (`--rsa` for an RSA key). The CN encodes `role.fqdn` so the role is the leading DNS label used by server ingest authorization (e.g. `CN=monitor.beryl.akoria.net`); the FQDN is also in the DNS SAN.
2. Submits `{token, role, fqdn, certFp, csrPem}` to the server — over HTTPS (`--server`) or a local `solariCtl` Unix socket (`--ctl-socket`).
3. Handles the `status=pending` reply (awaiting operator approval).
4. On approval, installs the signed cert + CA root, verifies the chain, and stamps `nodeId` into `[identity]` of the node config.

```
# enroll a monitor against the server over HTTPS
deploy/enrollment/solari-enroll.sh --role monitor --fqdn beryl.akoria.net \
    --token "$ENROLL_TOKEN" --server https://server.akoria.net

# generate key + CSR only (no submission) — useful for offline / staged enrollment
deploy/enrollment/solari-enroll.sh --role monitor --fqdn beryl.akoria.net --csr-only
```

Key flags: `--role`, `--fqdn`, `--token`, `--server` / `--ctl-socket`, `--rsa` / `--key-bits`, `--csr-only`, `--conf-dir` / `--config`, `--insecure`, `--force`.

> **Verified:** `--csr-only` produced a real EC key + CSR whose `openssl req -verify` returns "self-signature verify OK" (`subject=CN=monitor.beryl.akoria.net`).

Decommission runbook

The dashboard issues a durable decommission challenge on the first queued request. The second request supplies that same token; the server checks its stored digest, node, wipe scope, operator, and expiry before consuming it and publishing a directive with the challenge generation. The node remains active until an authenticated success result echoes that generation. Expired, mismatched, replayed, or force-cancelled challenges fail. The active server persists delivery attempts and retries from 5 seconds up to a 5-minute interval until acknowledgement or cancellation, including after restart or failover. The client currently has no receiver for this directive, so remote teardown still cannot complete. The privileged request queue also retains the raw confirmation token in its request/result text. Do not treat a sent directive as a completed wipe. `deploy/enrollment/solari-decommission.sh` remains a separate node-local, operator-driven executor (Handoff §4.2 / §7.3), not an SCP message receiver:

1. Requires a server-issued `--confirm-token` (`TLV_LIFE_CONFIRM`) echoed for audit, plus an interactive phrase confirmation.
2. `--wipe-scope config,certs,spool,logs,unit,all` (`TLV_LIFE_WIPE_SCOPE`) selects what is erased.
3. Stops the service, reports retirement best-effort *before* key deletion, securely erases the scoped material (`shred` when available), then disables / removes the unit.
4. `--dry-run` to rehearse; `--local-force --yes` for an offline or forced teardown when the server is unreachable.

```
# server-driven, audited teardown of a node
deploy/enrollment/solari-decommission.sh --confirm-token "$TOKEN" --wipe-scope all
# rehearse without touching anything
deploy/enrollment/solari-decommission.sh --confirm-token "$TOKEN" --wipe-scope config,certs
--dry-run
```

> **Verified:** scope parsing and a full `--dry-run` exit cleanly (rc=0); both scripts pass `bash -n` .

Retention & history rolling

`db/maintenance/` holds MariaDB stored procedures that operationalize the `TO_DAYS(sampledAt)` RANGE partition scheme from `db/schema.sql`:

- `retention.sql` — `solari_purge_retired` / `solari_purge_history` / `solari_retention_run`: purge `state='retired'` nodes (with FK-cascade children plus explicit `nodeConfig` / `hostHistory`) past the retention window, with a row-level history safety net.
- `roll_history.sql` — `solari_roll_partitions`: REORGANIZE the `p_max` catch-all into monthly `pYYYYMM` partitions ahead of now and DROP-PARTITION months aged past retention, on both `hostHistory` and `probeHistory`. Idempotent via `information_schema.PARTITIONS` guards; dynamic DDL via PREPARE/EXECUTE.

Schedule them from cron / an event scheduler, e.g. `CALL solari_retention_run(...)` nightly and `CALL solari_roll_partitions(...)` monthly.

CI

`.github/workflows/ci.yml` defines ten job groups and the current draft PR runs 15 hosted checks. The local `build-server-asan` CTest run passed **37/37**. The first hosted run passed 12 of 15 checks; subsequent runs caught a date-dependent SOA serial test, a missing RackWire mirror, a missing server-binary artifact in the dashboard job, server checkout omitting the RackWire submodule, and clean-install schema formatting across MariaDB 11.8 patch versions. The first four fixes passed in hosted CI; executable-comment normalization and failure diff output now have focused regression coverage and await a hosted rerun.

- `core` — the default zero-dependency build and CTest suite.
- `server` — full server build with ASan/UBSan, database migrations, enrollment and lease failover tests, scan-worker and adoption integration, and browser acceptance.
- `dashboard` — PHP/API and authorization tests, migration contracts, RackWire APIs, scanner-to-RackWire browser flow, and deployment-package checks.
- `rackwire-browser` — standalone offline, accessible, responsive planner flow.
- `python-integrations` — MCP, alertbridge, UniFi, authbroker, notifications, and Opie tests.
- `sorsync` — SoR outbox delivery across RabbitMQ releases and an upgrade/recovery rehearsal.
- `protocol-fuzz` — coverage-instrumented SCP parser fuzz smoke tests.
- `panel` — firmware, daemon, provisioning-tool, and dashboard parity suites.

- `cross` — Linux ARM, Windows, and other cross-target builds with qemu-user smoke runs where available.
- `macos` — universal build and native core tests.

Loopback transport staging is best-effort, mirroring `bootstrap-xenon.sh`. The RackWire and dashboard jobs check out submodules recursively; the RackWire GitHub mirror is available. Hosted validation remains in progress for the schema-export correction. Other CI results remain local or unverified as stated in their gates.

Known gaps & deferred items

> - **Production MariaDB adoption remains unverified.** Disposable MariaDB integration tests have exercised the migration ledger and selected SoR transaction paths. They do not prove migration adoption against current production databases, backups, or restore procedures. Review the current stabilization baseline before operational changes.
> - **Production browser / PHP-FPM acceptance remains open.** Authenticated browser tests now exercise multi-session freshness, scanner import, and embedded RackWire save against local PHP and disposable MariaDB fixtures. These do not establish behavior after deployment with PHP-FPM, production authentication, the live MariaDB, and `solariCtl`. The retired SSE endpoint is covered by a route test; a durable event bridge remains future work.
> - **Live SNMP device acceptance remains open.** Both LLDP topology fetching and IF-MIB interface polling are implemented behind `SOLARI_WITH_SNMP` and use the net-snmp CLI at runtime. Parser and merge tests pass, and the server builds with the feature both off and on. No live switch or router query has verified credentials, device OID support, data freshness, or persisted results in this release candidate.
> - **Production systemd and hosted CI acceptance remain open.** The section-placement check passes for 43 service files. Structural verification of 49 candidate unit files still returns 14 missing-executable diagnostics for binaries or virtual environments absent from this workspace. Enrollment and decommission scripts passed ShellCheck with sourced-file analysis. The GitHub workflow parses locally but has not run on GitHub from this session.

Status card updated 22 October 2026. Sections below preserve the Phase 7 implementation snapshot and its test counts. For current test results and release gates, use the linked `SolariNet_Stabilization_Baseline_2026-10-20` and

`SolariNet_Stabilization_Integration_Workflow_2026-08-09`. This guide does not declare the integration candidate ready for release.